

Network Analysis using Wireshark

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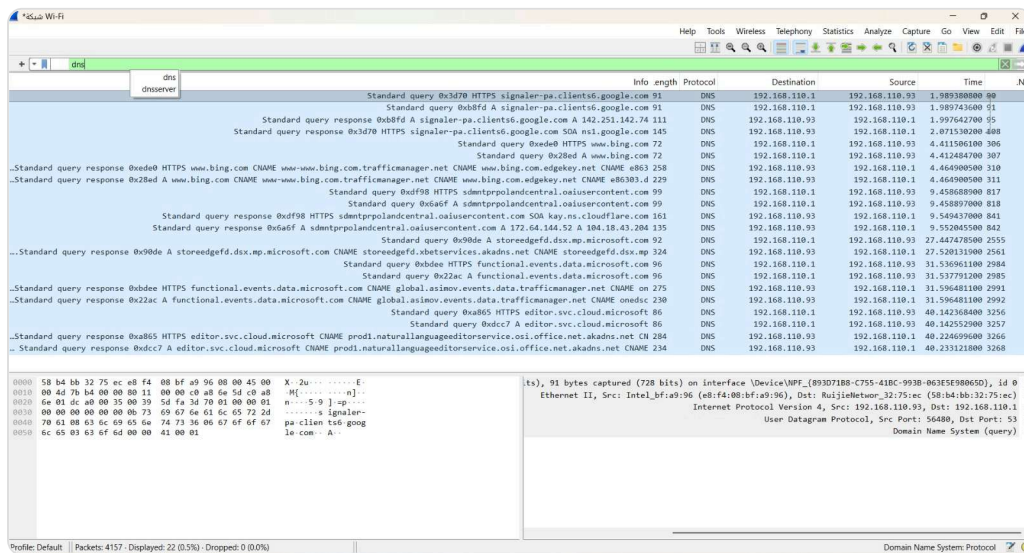
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Course: Networks

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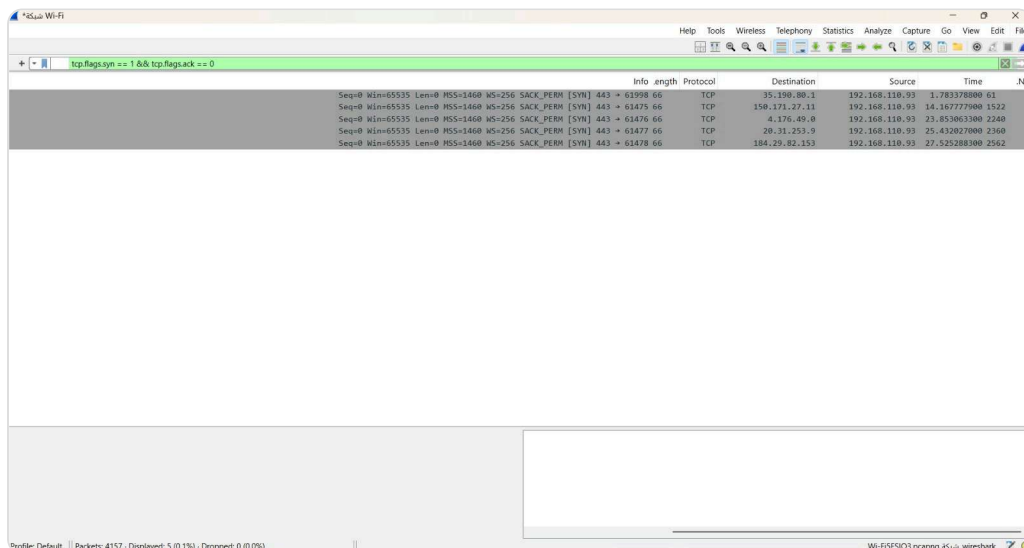
1. DNS

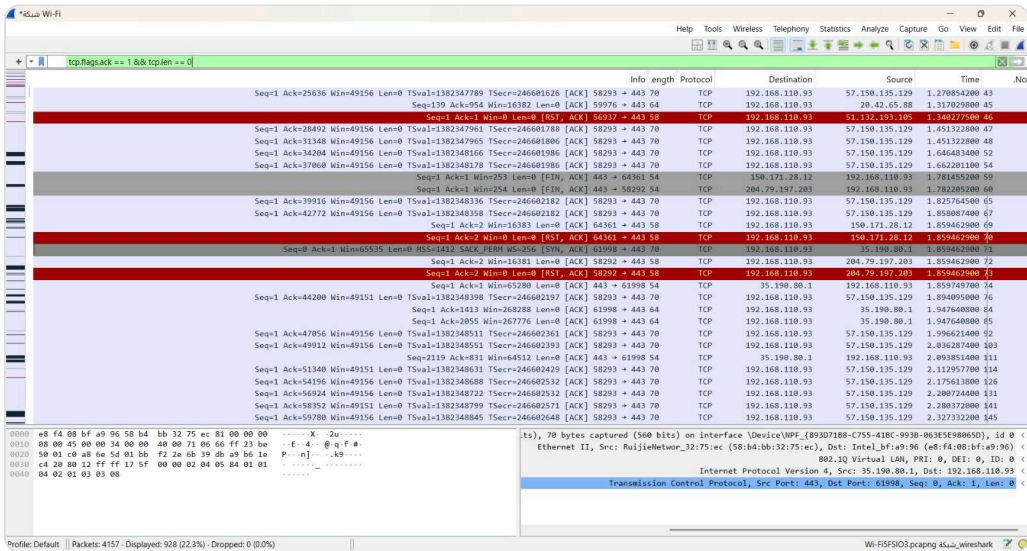
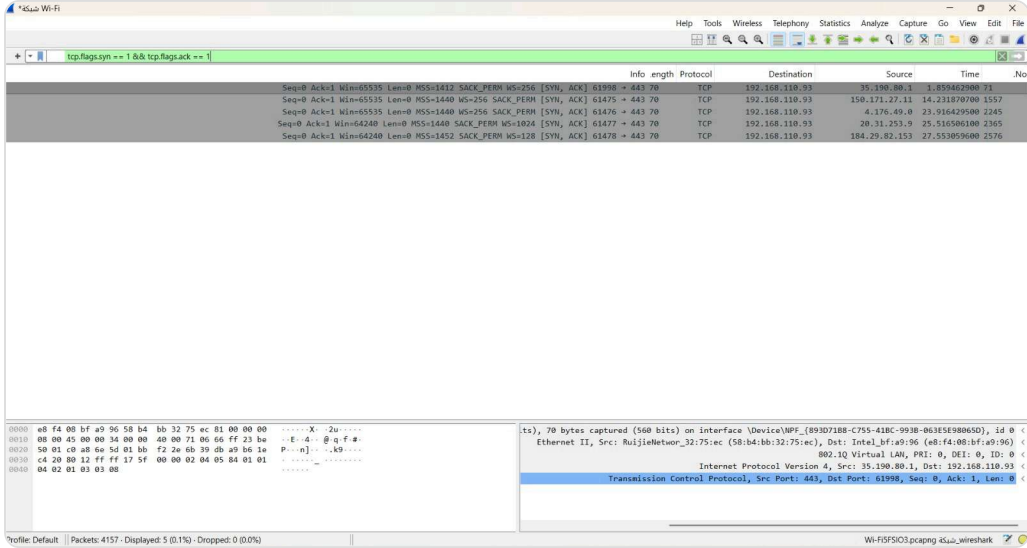
This shows a DNS query and response used to resolve a domain name.



2. TCP 3-Way Handshake

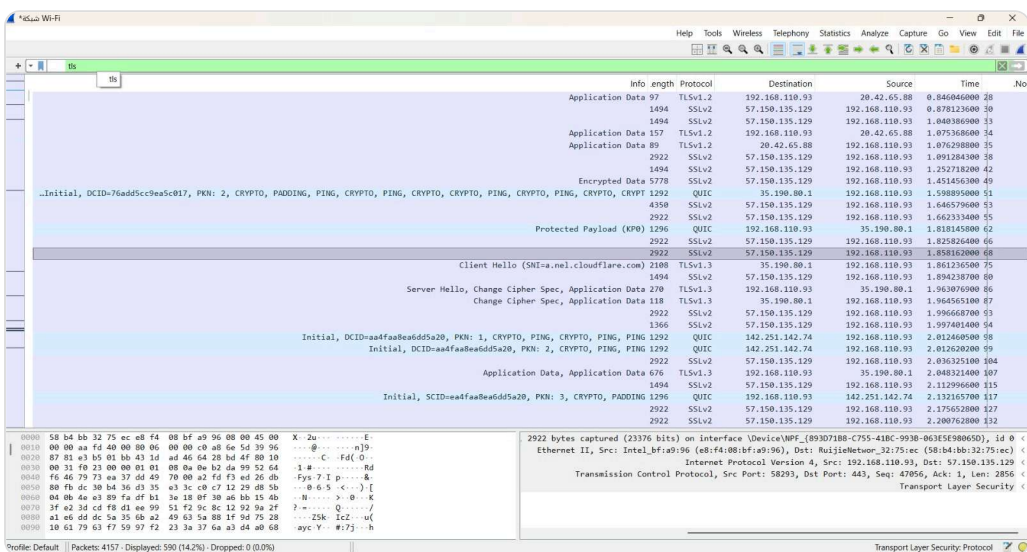
This shows SYN, SYN-ACK, and ACK packets.





3. TLS

This shows a secure HTTPS connection using TLS.



4. Results and Analysis

The captured traffic demonstrates the typical process of network communication. First, DNS queries are used to resolve domain names into IP addresses. Then, a TCP three-way handshake (SYN, SYN-ACK, ACK) establishes a reliable connection between client and server.

After the connection is established, TLS is used to secure the communication. The presence of Client Hello and Server Hello confirms the establishment of an encrypted HTTPS session.

RTT $\approx$ 0.07 seconds indicates normal network latency. Total packets captured $\approx$ 4157, showing active communication during the capture session.

TCP ensures reliability through acknowledgments (ACK), sequencing, and retransmission mechanisms, which guarantee correct and ordered delivery of data.

5. Conclusion

This analysis demonstrates how modern networks operate using DNS, TCP, and TLS. Wireshark provides deep visibility into network traffic and helps in understanding protocol behavior and troubleshooting network issues.